GeoAI Hackathon @ CSRE, IIT Bombay

Build an AI assistant (SLM) that recommends where and when to head outdoors

The challenge

Weather forecasts and maps tell you what the conditions will be. What people really want to know is more specific: which nearby place, at what time, is best for me, and what should I keep in mind before I go?

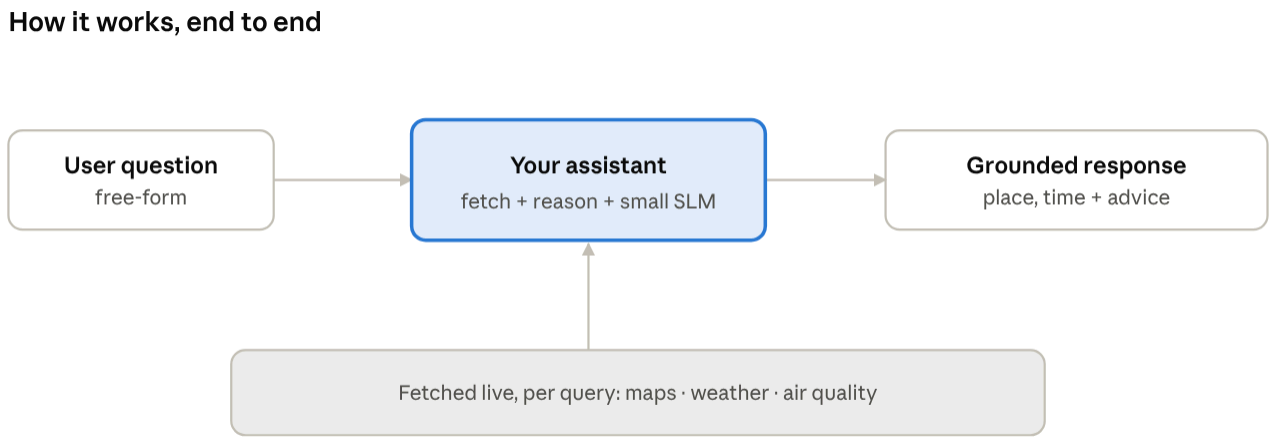
Build an **AI assistant (Small Language Model)**, powered by a small language model, that advises a person on an outdoor activity: the best nearby place and time to go, what to prepare, what to watch for, and when to avoid it. The assistant should reason over geospatial context and live weather to arrive at its recommendation.

Given a free-form question, the assistant should:

- identify the activity, the area or search radius, the time range, and any personal detail such as age or a health condition;
- consider several nearby places across the upcoming hours, rather than a single location;
- account for geospatial context, including wind direction, roads and traffic, nearby industry, water bodies and green cover;
- return a clear recommendation together with practical advice: what to carry, what to watch for, and when to avoid going;
- support every claim with the live data it actually used, without inventing weather or places;
- present the response in whatever form works best, whether plain text or something richer with maps and images.

What makes the problem interesting: the right answer depends on both the hour and the place. A ground that lies downwind of an expressway at 8 AM may be pleasant by evening once the wind shifts, and a venue forty minutes away at rush hour may not be worth the trip. The assistant has to reason about place and time together, for this particular person.

At a glance



Nothing is supplied in advance. For each question, the assistant fetches what it needs in real time, reasons about place and time using a small language model, and returns a grounded recommendation with supporting advice.

Activities to support

Treat the following as your reference set. Your assistant should support **at least these ten** outdoor activities:

- Running or jogging
- Cycling (leisure or commute)
- Golf
- Cricket (recreational match or practice)
- Outdoor football or five-a-side
- Outdoor tennis
- Hiking or a nature walk
- Picnic or park outing
- Outdoor event or gathering (for example a wedding or fair)
- Outdoor photography session

Each activity has its own sensitivities to weather, air quality and daylight, so the same conditions can suit one and rule out another. Personal profiles, such as asthma, a senior, a child or pregnancy, should tighten the limits further for that user.

Example Queries

The following are the kinds of free-form questions your assistant should handle. They illustrate the input only; how you shape the response is up to you.

- "Where can I go running tomorrow evening near Koregaon Park?"
- "Best time this weekend for a kids' cricket match within 15 km of Whitefield?"
- "I have mild asthma, where's good for a morning walk near Powai in the next two days?"
- "Is it okay to cycle to work at 8 AM?"
- "Planning an outdoor birthday party on Sunday afternoon in Pune, what should I keep in mind?"
- "Good place and time for golf on Saturday within 40 km of Andheri?"
- "Where can I take sunset photos this evening around Bandra?"
- "Is Saturday or Sunday better for a hike near Lonavala?"
- "When is it safe for my parents to walk outdoors tomorrow in Bengaluru?"
- "Outdoor tennis after work today near Hinjewadi, any good spots?"

Expect incomplete or messy questions as well: a missing location, a vague time such as "this weekend", or a request that falls outside the scope. Handle these gracefully rather than guessing. **This is just for your understanding the evaluation queries would be different then these.**

Data: fetch it live

No dataset is provided. We supply no map files, no weather files and no server. Your assistant must fetch everything it needs **live, at query time**, from free and open sources. Sourcing and handling that data is part of the challenge, since a real assistant has to do the same.

The sources below are a starting point, not a requirement. You may use any free source, provided you **disclose** it. Respect each service's rate limits, and cache responses sensibly.

Maps, places and features (venues, roads, industrial zones, water, parks, power lines)

- OpenStreetMap via the **Overpass API.**
- **Geofabrik:** downloadable regional OSM extracts
- **Nominatim:** geocoding (place name - coordinates)

Weather and air quality

- **Open-Meteo:** forecast + air-quality APIs
- **OpenAQ:** open air-quality measurements
- **WAQI:** real-time AQI, free token
- **OpenWeatherMap:** free tier, needs a key (alternative)

Routing and Travel Time

- **OSRM or OpenRouteservice**

Daylight and terrain

- **Sunrise-sunset API, Open-Elevation/ OpenTopoData**

Indian Open data

- **Data.gov.in and CPCP for air quality and IMD for weather**

You are free to use any other free and open source beyond those listed. Whatever you use, you must explicitly state your data sources.

Your model: a small open SLM on Hugging Face

The language component of your assistant must be a **small, open-weights model of at most 3 billion parameters** that you run yourself. A small model cannot memorise the world's weather and geography, and that limitation is the point: it forces the system to rely on live data rather than recall.

You are not expected to train a model from scratch; pretraining is beyond the scope of a hackathon. You may use an open base model directly (for example Phi-3-mini, Llama-3.2, Qwen2.5 or Gemma-2), or fine-tune one (for example with LoRA or QLoRA). Either approach is acceptable, and the choice is yours.

Host your model on the Hugging Face Hub as a public repository. Include the weights (or your LoRA adapter together with a pointer to the exact base model), a model card describing the base model, your changes and intended use, and an open licence. It should load with from_pretrained(...). If you use a base model unchanged, still create a repository with your configuration and a model card naming the base model and how you use it.

Large hosted models, such as GPT-4-class systems, Claude or Gemini Pro, may **not** be used within the running assistant.

Deliverables

Each team submits the following:

1. **A live API endpoint:** an HTTP endpoint we can call to send questions to your assistant, kept available throughout the judging window.
2. **Your model on Hugging Face** — a public repository with the model or adapter, a model card and an open licence.
3. **A code repository:** your complete source code.
4. **A short README:** how to run the project, the base model you used, the data sources you relied on, and your API endpoint URL.
5. **A presentation (4–6 slides):** the problem and your approach, your architecture, a worked example, your model and any fine-tuning, and an honest note on limitations.

Be ready to demonstrate your assistant live during judging.

Submission

Submissions will be collected through a Google Form. The link will be made available soon.

Rules

Allowed

- Any open-weights language model of at most 3B parameters (disclose the exact model).
- Fine-tuning that model (LoRA / QLoRA or similar).
- Any free, openly available data source or API — each disclosed.
- Any open-source libraries, and AI coding assistants for writing code (disclosed).

Not allowed

- Large hosted models (GPT-4-class, Claude, Gemini Pro) anywhere in the running assistant.
- Paid or private data or APIs; scraping behind logins.
- Hard-coding answers to the example questions.
- Answering from the model's memory instead of the data it fetched.

Disclose the models, data sources and tools you used, and be prepared to explain and defend your work during judging.

Timeline and logistics

Key details for the event:

- **Mode** - online.
- **Submission deadline** - 24 October, 11:59 PM IST.
- **Finals** - The top five teams will be invited to IIT Bombay to present at the conference.

FAQ

Do we have to train a model from scratch? No. Use an open model of at most 3 billion parameters. Fine-tuning is optional, and pretraining is not expected.

Can we use ChatGPT, Claude or Gemini? Not within the running assistant. You may use them as coding aids while you build, provided you disclose it.

Where does the data come from? You fetch it live from free, open APIs, listed under Data. No dataset is provided.

Do we need a GPU? Not to run the assistant. If you choose to fine-tune, a free Colab or Kaggle GPU is sufficient.

What if our API endpoint is unavailable during judging? Keep it live for the full judging window. If we cannot reach it, we will rely on your code, README and demo, which is less ideal, so test your hosting early.

Can we support more than ten activities? Yes. Ten is the minimum; additional, well-handled activities are welcome.

What is the team size? To be announced.

Attribution and sources

If you use OpenStreetMap data, you must credit it as “© OpenStreetMap contributors”. Review and respect the terms and rate limits of every source you use.

- OpenStreetMap · Overpass API · Nominatim
- Open-Meteo · OpenAQ · WAQI
- OpenRouteService · OSRM · Sunrise-Sunset API
- Hugging Face Hub for hosting your model